REVA UNIVERSITY
Bengaluru, India

School of Computer Science & Engineering
AI Application Development – IA1

Product Name:
Deforestation Detection and Alert System using remote sensing images & vision models

Team members & roles

Team member name	Role
Prajwal R H	Developer Lead
Prajwal S A	Team Lead
Prem N Gowda	Customer Lead
Pruthviraj DK	Test Lead
Ravitej Achanur	Developer
Sai Sharan	Tester

User/ Customer Understanding

Modified Problem Statement

The 3 Core Problems

Problem	What Officers Said	Hard Number
Monsoon Blindness	"4 months completely blind, mafia rules"	33% year unmonitored
No Mobile/Offline	"Websites don't load in forest, no signal"	0% field accessibility
No Cause Classification	"Just says 'forest loss' — can't file right FIR"	0 systems classify cause

The Impact

Metric	Value
Annual forest loss	**150,000 hectares**
Daily loss	**400 hectares**
Hourly loss	**17 hectares**
Field officers with tools	**0 out of 50,000**

Problem Statement

India's 50,000+ forest officers have **NO monsoon monitoring**, **NO mobile alerts**, and **NO cause classification** — resulting in 150,000 hectares lost annually while existing systems fail.

Ideal Customer Profile (ICP)

Demographic Profile

Customer Type	Count	Area Managed	Current Tools
Beat Guards	35,000+	10-30 sq km each	ZERO
Range Officers	12,000+	100-300 sq km each	ZERO
Foresters	8,000+	50-100 sq km each	ZERO

Psychographic Profile — What They NEED

Need	Current Reality	What They Want
Monsoon Vision	Blind 4 months (optical fails)	SAR radar that sees through clouds
Mobile Alerts	Desktop-only, English-only	SMS/WhatsApp in local language
Offline Access	Need internet to use	Works with no signal
Cause Info	"Forest loss detected"	"Illegal logging detected"

Key Insight : **0% of beat guards** have EVER received an automated deforestation alert in their career.

Customer Persona (CP)

Demographic Profile

Attribute	Detail
Name	Ramesh Kumar (Representative Persona)
Age	42 years
Role	Range Forest Officer (RFO)
Location	Bandipur, Karnataka
Experience	15 years
Area Managed	180 sq km
Team	1 Forester + 8 Beat Guards
Phone	₹12,000 Redmi (Android)
Internet	2G in forest, 4G only in town
Language	Kannada (primary), Hindi (secondary)

Problem	Ramesh's Words
Monsoon	*"June to September, I'm completely blind. Smugglers know this — it's their peak season."*
No Mobile	*"Global Forest Watch doesn't load on my phone. In forest, I have no signal anyway."*
No Cause	*"Reports say 'forest loss' but was it logging or mining? Penalties are different. I can't file right case."*

Requirement	His Words
Works in monsoon	*"I need eyes that work even when it rains"*
Works offline	*"If it needs internet, it's useless in my range"*
Tells WHY	*"Tell me logging, mining, or fire — so I file correct FIR"*
Simple	*"Make it like WhatsApp — my guards will use it"*

Metric	Before DeforestNet	After DeforestNet
Monsoon monitoring	0%	100%
Alert access	Never	Direct on phone
Cause identification	Go to site manually	Instant classification

Product USP

Monsoon Monitoring (SAR Radar)

Problem: "4 months blind"
Solution: Radio waves penetrate clouds
Benefit: 100% year-round vs 0% (competitors)

Offline Mobile (Q9)

Problem: "No internet in forest"
Solution: SMS + WhatsApp + Offline + 6 languages
Benefit: Works on ₹8,000 phones

Cause Classification (Q5)

Problem: "Can't tell logging vs mining"
Solution: 5-class AI + recommended FIR
Benefit: Correct legal action, right penalty

Use Cases & High level Requirements

UC 1: All-Weather Detection

Input: Sentinel-1 + Sentinel-2
Output: Deforestation mask
Benefit: 365-day monitoring

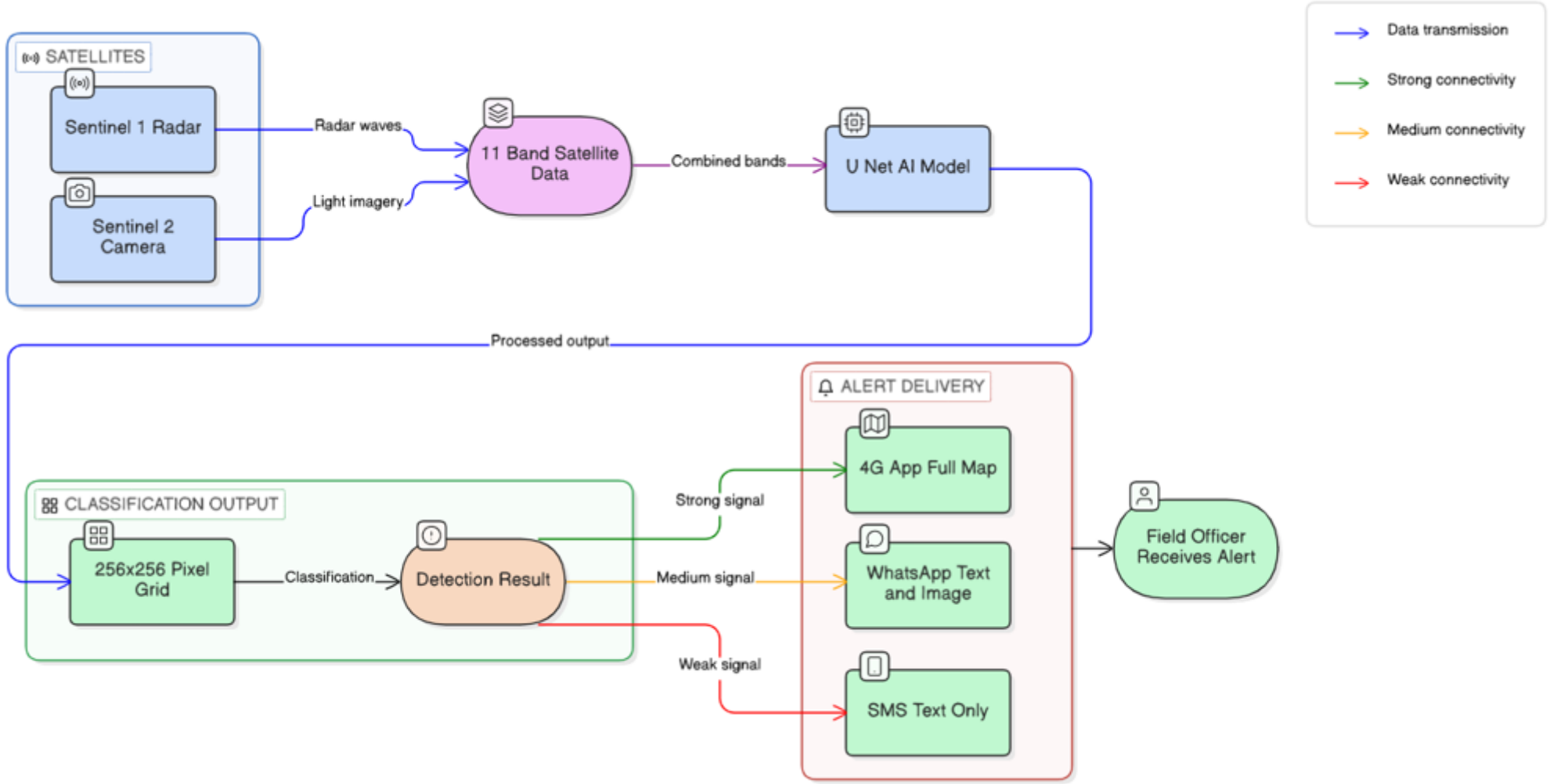
UC 2: Monsoon Monitoring

Input: SAR radar only
Output: Alerts in heavy rain
Benefit: No blind months

UC 3: Mobile Alerts

Input: Detection + GPS
Output: SMS/WhatsApp/App
Benefit: Reaches field officers in 5 days

UC 4: Cause Classification

Input: Deforestation patch
Output: Logging/Mining/Fire etc + FIR type
Benefit: Right case, right penalty

Proof of Concept (PoC) / Prototype Demo